

EASY SQL NOTES

DDL • DML • TCL | Simple, exam-friendly revision

1. DDL — Data Definition Language

Meaning: DDL is used to create or change the **structure** of database objects such as tables.

Command	Easy meaning	Example
CREATE	Creates a new table or database object	CREATE TABLE Student (id INT, name VARCHAR(30));
ALTER	Changes an existing table structure	ALTER TABLE Student ADD age INT;
DROP	Deletes the table/object completely	DROP TABLE Student;
TRUNCATE	Removes all rows but keeps the table structure	TRUNCATE TABLE Student;

Quick idea: Think: **DDL = Design the structure.**

2. DML — Data Manipulation Language

Meaning: DML is used to add, change, remove and retrieve the **data/records** stored in tables.

Command	Easy meaning	Example
INSERT	Adds new records	INSERT INTO Student VALUES (1,'Priya');
UPDATE	Changes existing records	UPDATE Student SET name='Anu' WHERE id=1;
DELETE	Removes selected records	DELETE FROM Student WHERE id=1;
SELECT	Displays/retrieves data	SELECT * FROM Student;

Quick idea: Think: **DML = Manage the data.**

Simple flow

TABLE

↓

INSERT → UPDATE → DELETE → SELECT

3. TCL — Transaction Control Language

Meaning: TCL controls transactions and decides whether database changes should be saved or undone.

Command	Easy meaning
COMMIT	Permanently saves the changes.
ROLLBACK	Undoes changes made after the last COMMIT.
SAVEPOINT	Creates a point for partial rollback.

Example:

```
UPDATE Student SET name='Priya' WHERE id=1;
SAVEPOINT s1;
UPDATE Student SET name='Anu' WHERE id=1;
ROLLBACK TO s1;
COMMIT;
```

The SAVEPOINT lets us return to an earlier safe point. COMMIT saves the final accepted changes.

QUICK COMPARISON

DDL	DML	TCL
Structure	Data	Transactions
CREATE ALTER DROP TRUNCATE	INSERT UPDATE DELETE SELECT	COMMIT ROLLBACK SAVEPOINT
Build / change the table	Work with the records	Save / undo the changes

■ EASY MEMORY TRICK

DDL	D = Design → table structure
DML	M = Manage → table data
TCL	T = Transaction → save or undo

One-line revision: DDL creates the structure, DML handles the data, and TCL controls the transaction.

Exam tip: For a short answer, write the full form + definition + 2–3 commands + one example.